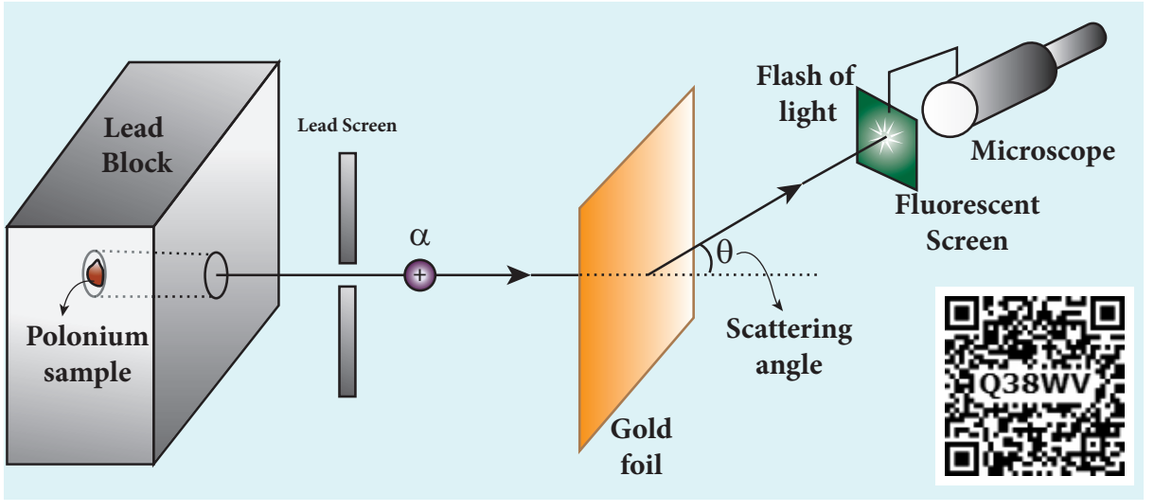


Figure 9.9 Schematic diagram for scattering of alpha particles experiment by Rutherford

kept inside a thick lead box with a fine hole as seen in Figure 9.9. The alpha particles coming through the fine hole of lead box pass through another fine hole made on the lead screen. These particles are now allowed to fall on a thin gold foil and it is observed that the alpha particles passing through gold foil are scattered through different angles. A movable screen (from 0° to 180°) which is made up of zinc sulphide (ZnS) is kept on the other side of the gold foil to collect the scattered alpha particles. Whenever alpha particles strike the screen, a flash of light is observed which can be seen through a microscope.

Rutherford proposed an atom model based on the results of alpha scattering experiment. In this experiment, alpha particles (positively charged particles) were allowed to fall on the atoms of a metallic gold foil. The results of this experiment are given below and are shown in Figure 9.10, Rutherford expected the atom model to be as seen in Figure 9.10 (a) but the experiment showed the model as in Figure 9.10 (b).

(a) Most of the alpha particles were un-deflected through the gold foil and went straight.

(b) Some of the alpha particles were deflected through a small angle.

(c) A few alpha particles (one in thousand) were deflected through the angle more than 90°

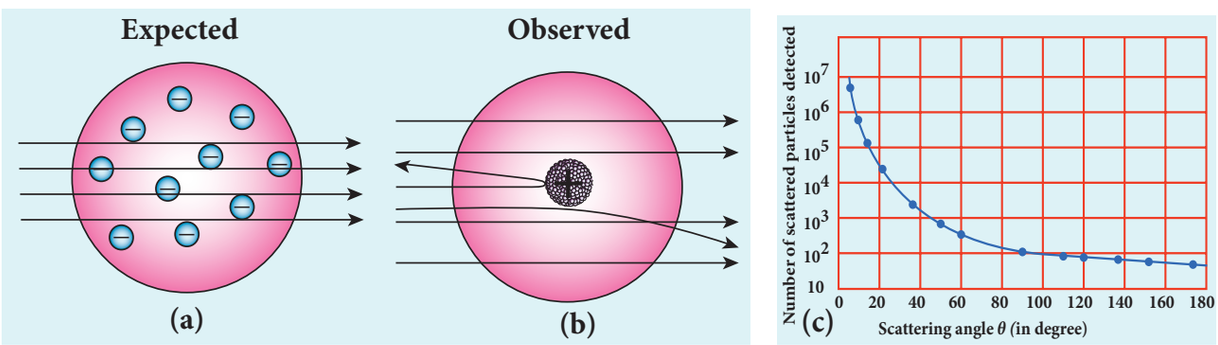


Figure 9.10 In alpha scattering experiment – (a) Rutherford expected (b) experiment result (c) The variation of alpha particles scattered N(θ) with scattering angle θ

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